CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EE) Examination (Backlog)

Nov 2012

EE 204 Electrical Power System – I

Date: 06.11.2012, Tuesday | **Time: 01:30 p.m. To 04:30 p.m.** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Derive the equation of the inductance of three phase transmission line [07]

Q - 2 (a) A 400 kV, 3 – phase bundled conductor line with two sub conductor per phase has a horizontal configuration as shown in Figure 1. The radius of each sub-conductor is 1.6 cm. (a) Find the inductance per phase per km of line (b) Compute the inductance of the line with only one conductor per phase having the same cross sectional area of the conductor of each phase. [14]

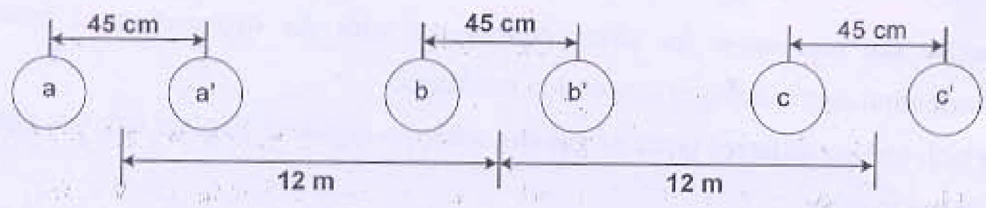

Figure 1

(b) Write a short note on bundled conductors and state the advantages of it

(c) Explain the transposition in transmission line? Why is it required?

OR

Q - 2 (a) Derive the equation to find the capacitance of three phase line with asymmetrical spacing of the conductors. [14]

(b) For a 220 kV, 50 Hz, 200 km long three phase line, the distance between a to b, b to c and a to c are 6 m , 6 m and 12 m, respectively. The conductor radius is 1.81 cm. find the capacitance per phase per km, capacitive reactance per phase, charging current and total charging mega-voltamperes.

(c) What is skin and proximity effect in transmission lines

Q - 3 (a) Derive the ABCD parameters for medium transmission line considering (a) nominal T configuration and (b) nominal pi configuration [14]

(b) A 15 km, 3 – phase overhead transmission line delivers 5 MW at 11 kV at a power factor of 0.8 lagging. Line loss is 12 % of the power delivered. Line inductance is 1.1 mH per km per phase. Calculate (a) sending end voltage and voltage regulation (b) power factor of the load to make regulation zero.

(c) Explain the Ferranti effect in transmission line. Draw the voltage phasor diagram for transmission network under Ferranti effect.

OR

Q - 3 (a) For the long transmission line, derive the relationship between sending end and receiving end voltage and current. [14]

(b) A 132 kV, 3 – phase, 50 Hz transmission line has the following parameters: R = 0.11 ohm/km, L = 1.5 mH/km and c = 0.01 microF/km. The transmission line is 150 km long and is supplying a lagging load of 50 MW at 0.8 PF at 125 kV. Determine the sending end voltage.

(c) Explain the tuned power lines in power system.

SECTION – II

Q - 4 (a) Derive the expression for string considering with the explanation of potential distribution over a string of suspension insulators. [07]

Q - 5 (a) Which are the different types of grading schemes in cable? Explain any one grading scheme in cable. [14]

(b) A conductor of 1 cm diameter passes centrally through a porcelain cylinder of internal diameter 2 cms and external diameter 7 cms. The cylinder is surrounded by a tightly fitting metal sheath. The permittivity of porcelain is 5 and the peak voltage gradient in air must not exceed 34 kV/cm. Determine the maximum safe voltage.

OR

Q - 5 (a) List out the different types of tariff and explain the following in detail: [14]

(a) two part tariff (b) three part tariff and (c) power factor tariff

(b) A consumer takes a steady load of 250 kW at a power factor of 0.8 lagging for 10 hrs per day and 300 days per annum. Estimate the annual payment under each of the following tariff:

(i) Rs. 1.20 per kWh + Rs. 1200 per kVA per annum

(ii) Rs. 1.20 per kWh + Rs. 1200 per kW per annum + 25 paise per kVARh

Q - 6 (a) What are the causes of low power factor? Discuss the different methods for power factor improvement. [14]

(b) Derive the equation for most economical power factor.

OR

Q - 6 (a) Define the following terms related to economic operation of power system. [14]

(a) load curve (b) load duration curve (c) load factor (d) diversity factor (e) demand factor (f) plant factor (g) maximum demand

(b) The load on a power plant on a typical day is as under:

Time	12 – 5 AM	5 – 9 AM	9 – 6 PM	6 – 10 PM	10 – 12AM
Load (MW)	20	40	80	100	20

Plot the load curve and load duration curve. Find the load factor of the plant and energy supplied by the plant in 24 Hrs.

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